

CSE2104: PROGRAMMING LANGUAGE LAB

Lab Task Sheet — Array and Functional Problems

SL	Task				
1	Write a program to find the largest element in an array. The user will input the size of the array and its elements. INPUT: • The first line contains an integer n ($1 \leq n \leq 100$), the size of the array. • The second line contains n integers separated by spaces, representing the elements of the array. OUTPUT: • A single integer: the largest element in the array. <table><tr><th>Sample Input</th><th>Sample Output</th></tr><tr><td>5 10 2 30 14 25</td><td>30</td></tr></table>	Sample Input	Sample Output	5 10 2 30 14 25	30
Sample Input	Sample Output				
5 10 2 30 14 25	30				
2	Write a program to reverse the elements of an array. The user will input the size of the array and its elements. INPUT: • The first line contains an integer n ($1 \leq n \leq 100$), the size of the array. • The second line contains n integers separated by spaces, representing the elements of the array. OUTPUT: • A single line containing the reversed array. <table><tr><th>Sample Input</th><th>Sample Output</th></tr><tr><td>5 10 2 30 14 25</td><td>25 14 30 2 10</td></tr></table>	Sample Input	Sample Output	5 10 2 30 14 25	25 14 30 2 10
Sample Input	Sample Output				
5 10 2 30 14 25	25 14 30 2 10				
3	Write a C program to count how many odd digits are present in a given number. <table><tr><th>Sample Input</th><th>Sample Output</th></tr><tr><td>24679</td><td>2</td></tr></table>	Sample Input	Sample Output	24679	2
Sample Input	Sample Output				
24679	2				

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4	In an HR system, employee IDs are encrypted by adding 5 to each digit of the ID and returning the new number. Write a function to encrypt the employee ID using this logic. If any digit exceeds 9 after addition, take only the last digit (e.g., 13 becomes 3). The input contains a positive integer representing the employee ID. <table> <tr> <th>Sample Input</th><th>Sample Output</th></tr> <tr> <td>Employee ID: 4823</td><td>9378</td></tr> </table>	Sample Input	Sample Output	Employee ID: 4823	9378
Sample Input	Sample Output				
Employee ID: 4823	9378				
5	John is working on a weather monitoring system that records temperatures in an array. He wants to shift the last K days temperatures to the front while pushing the rest back. Write a C program that rotates the array to the right by K positions. <table> <tr> <th>Sample Input</th><th>Sample Output</th></tr> <tr> <td> Array size: 6 Array elements: 10 20 30 40 50 60 K: 2 </td><td>50 60 10 20 30 40</td></tr> </table>	Sample Input	Sample Output	Array size: 6 Array elements: 10 20 30 40 50 60 K: 2	50 60 10 20 30 40
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